

Shift-Based Augmented Coprime Array with Reduced Mutual Coupling Effect

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Abstract—In this paper, sparse array design with increased degrees of freedom (DOFs) and reduced mutual coupling is investigated. Shift-based augmented coprime array (SACA), which is generated by strategically relocating the last sensor of the subarray with element spacing M in the original augmented coprime array (ACA) structure, is proposed. The number of uniform degrees of freedom (uDOFs) of the SACA is proved to be the same as ACA, while reduced critical weight functions can be achieved. As shown by simulations, the proposed SACA exhibits improved direction-of-arrival (DOA) estimation performance compared with ACA and other existing structures.

Index Terms—Sparse array design, difference co-array, coprime array, direction of arrival estimation, mutual coupling.

I. INTRODUCTION

DOA estimation with sensor arrays is a classical problem in the field of signal processing, widely applied to areas such as radar, communications, navigation, and sonar [1]–[5]. Uniform linear array (ULA) is the most fundamental and commonly used array configuration, where up to $N - 1$ sources can be distinguished based on an N -element ULA [6], [7]. Compared to ULA, sparse array enables resolving more sources than the number of physical sensors, making underdetermined DOA estimation feasible, and thus have been extensively studied [8]–[11]. Minimum redundancy array (MRA) [12] and minimum hole array (MHA) [13] are two examples of sparse array; however, a key drawback is the lack of closed-form expression for their sensor positions.

Then, nested array (NA) [14], coprime array (CA) [15], and their various improved forms [16]–[20] have been proposed. NA has a closed-form expression for its sensor positions and offers a high degrees of freedom (DOFs). However, its dense sub-array introduces strong mutual coupling (MC) effect. To reduce the MC effect while maintaining the number of DOFs provided, super nested array (SNA) [21] is proposed, reducing the number of sensor pairs with small inter-element spacings. AMDNA [22] is obtained by redistributing the dense uniform linear subarray in NA, with reduced MC achieved.

Compared to NA, CA exhibits a stronger robustness against MC effect. However, holes exist in the difference co-array of CA, disrupting the continuity of the entire virtual co-array, and the number of resolvable sources for CA is relatively limited

when subspace-based methods are employed. Thinned coprime array (TCA) [23] reduces redundant sensors in the original CA, achieving the same virtual aperture and uniform DOFs (uDOFs) as the original CA with fewer physical sensors. ACA [24] is capable of achieving the same uDOF as NA with the same number of sensors while maintaining robustness against MC comparable to CA. In [25], the maximum inter-element spacing constraint (MISC) array is introduced, and increased uDOFs is provided with reduced coupling leakage.

In this paper, aiming at further reducing the MC effect while maintaining a high number of uDOFs, we propose a sparse array design, referred to as the shift-based augmented coprime array (SACA). By analyzing the hole locations and hole-filling strategy in ACA, SACA configuration is obtained by relocating the last sensor of the sub-array with inter-element spacing M in the original ACA. Analysis reveals that the proposed SACA and ACA provide the same number of uDOFs. Moreover, this sensor relocation enables a reduction in the first three critical weight functions for certain sensor numbers, which in turn suppresses the MC effect. As verified by the simulations, superior estimation performance can be achieved.

The rest of this paper is organized as follows. The signal model is introduced in Section II. The proposed SACA structure is presented in Section III together with analysis of uDOFs and weight functions. Simulation results are given in Section IV, while conclusions are drawn in Section V.

II. SIGNAL MODEL

Consider a linear array with N sensors, and the set of sensor positions is

$$\mathbb{P} = \{p_n d \mid n = 0, 1, \dots, N - 1\}, \quad (1)$$

where $p_n d$ denotes the position of the n th sensor, and d is the unit spacing which is omitted hereafter for simplification.

Assume that K uncorrelated narrowband far-field sources impinge on the linear array from directions θ_1 to θ_K . The received signal model is expressed as

$$\mathbf{x}(i) = \mathbf{A}(\theta)\mathbf{s}(i) + \mathbf{n}(i), \quad (2)$$

where $\mathbf{A}(\theta) = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_K)]$ denotes the array steering matrix, $\mathbf{s}(i)$ denotes the source signal vector, and $\mathbf{n}(i)$ represents the additive noise vector.

The covariance matrix of $\mathbf{x}(i)$ is

$$\begin{aligned} \mathbf{R} &= \mathbb{E} [\mathbf{x}(i)\mathbf{x}^H(i)] = \sum_{k=1}^K \sigma_k^2 \mathbf{a}(\theta_k) \mathbf{a}^H(\theta_k) + \sigma_n^2 \mathbf{I} \\ &= \mathbf{A} \mathbf{R}_s \mathbf{A}^H + \sigma_n^2 \mathbf{I}, \end{aligned} \quad (3)$$

where $\mathbf{R}_s = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_K^2)$ denotes the source covariance matrix with $\text{diag}(\cdot)$ returning a diagonal matrix, and σ_n^2 is the noise power.

Vectorizing the covariance matrix $\mathbf{R}$, we obtain

$$\mathbf{r} = \text{vec}(\mathbf{R}) = \mathbf{A}_d \mathbf{p} + \sigma_n^2 \mathbf{i}, \quad (4)$$

where $\mathbf{A}_d$ represents the equivalent steering matrix of the difference co-array [14] [17], $\mathbf{A}_d = \mathbf{A}^* \odot \mathbf{A}$ with $\odot$ denoting the Khatri-Rao product, $\mathbf{p} = [p_1, p_2, \dots, p_K]^T$, and $\mathbf{i} = \text{vec}(\mathbf{I})$.

In the presence of mutual coupling, the received signal model is changed to

$$\mathbf{x}(i) = \mathbf{C} \mathbf{A} \mathbf{s}(i) + \mathbf{n}(i). \quad (5)$$

Generally, $\mathbf{C}$ is a B -banded MC matrix with its (i, j) -th entry given by [21]

$$C_{i,j} = \begin{cases} c_{|p_i - p_j|}, & |p_i - p_j| \leq B, \\ 0, & \text{otherwise}, \end{cases} \quad (6)$$

where c_i is the coupling coefficient for two sensors with a distance of i , satisfying $1 = |c_0| > |c_1| > \dots > |c_B| > |c_{B+1}| = 0$.

The coupling leakage is defined as [24] [26]

$$\mathcal{L} = \frac{\|\mathbf{C} - \text{diag}(\mathbf{C})\|_F}{\|\mathbf{C}\|_F}. \quad (7)$$

where $\|\mathbf{C} - \text{diag}(\mathbf{C})\|_F$ is the energy of the off-diagonal entries, which characterizes the impact of mutual coupling.

III. PROPOSED ARRAY GEOMETRY

A. Shift-Based Augmented Coprime Array

We aim at reducing MC effect of the array without compromising the number of uDOFs provided by ACA, thereby achieving better DOA estimation performance. By shifting the last sensor of the subarray with spacing M in ACA from position $MN - M$ to $MN + N$, the shift-based augmented coprime array, referred to as SACA, is obtained.

Definition 1: The SACA structure comprises four subarrays, and its sensor position set is

$$\mathbb{P} = \mathbb{P}_1 \cup \mathbb{P}_2 \cup \mathbb{P}_3 \cup \mathbb{P}_4, \quad (8)$$

where

$$\begin{cases} \mathbb{P}_1 = \{mN \mid m \in [0, M-1]\}, \\ \mathbb{P}_2 = \{nM \mid n \in [0, N-2]\}, \\ \mathbb{P}_3 = \{-q(M+N) \mid q \in [1, Q]\}, \\ \mathbb{P}_4 = \{MN, MN+N\}. \end{cases} \quad (9)$$

with M and N are coprime integers, and $N > M \geq 2$.

The array geometry of SACA for $M = 2$, $N = 3$, and $Q = 3$ is illustrated in Fig. 1. Then, the number of uDOFs provided by the proposed SACA is proved in the following proposition.

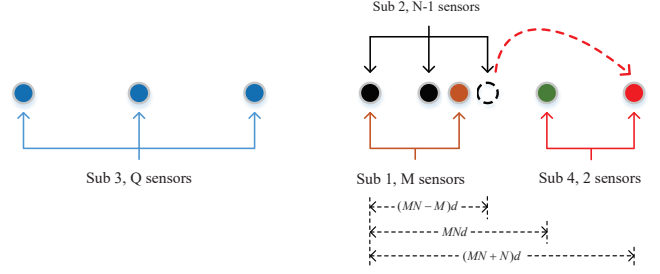


Fig. 1: Geometry of SACA for $M = 2$, $N = 3$, and $Q = 3$.

Proposition 1: For a given number of sensors, SACA configuration offers the same number of uDOFs as the ACA structure, which is $2(Q+1)(M+N)-1$.

Proof: For the ACA configuration, denote $\mathbb{P}_1 \cup \mathbb{P}_2 \cup (MN - M) \cup (MN) = \hat{\mathbb{V}}_1 \cup \hat{\mathbb{V}}_2$, where

$$\begin{cases} \hat{\mathbb{V}}_1 = \{m'N \mid m' \in [0, M]\}, \\ \hat{\mathbb{V}}_2 = \{n'M \mid n' \in [0, N]\}. \end{cases} \quad (10)$$

Here, $\hat{\mathbb{V}}_1 \cup \hat{\mathbb{V}}_2$ is the set composed of the first two subarrays of ACA and the sensor at MN .

Subsequently, for SACA configuration, denote $\mathbb{P}_1 \cup \mathbb{P}_2 \cup (MN) = \mathbb{V}_1 \cup \mathbb{V}_2$ with

$$\begin{cases} \mathbb{V}_1 = \{m'N \mid m' \in [0, M]\}, \\ \mathbb{V}_2 = \{n''M \mid n'' \in [0, N] \setminus \{N-1\}\}, \end{cases} \quad (11)$$

where $\mathbb{A} \setminus \mathbb{B}$ denotes the relative complement of $\mathbb{B}$ in $\mathbb{A}$, and $\mathbb{V}_1 \cup \mathbb{V}_2$ represents the set composed of the first two subarrays of SACA and the sensor at MN .

Firstly, we prove that from ACA to SACA, removing the sensor at position $MN - M$ will not change the difference co-array generated from subarrays $\hat{\mathbb{V}}_1$ and $\hat{\mathbb{V}}_2$.

For $\hat{\mathbb{V}}_1 \cup \hat{\mathbb{V}}_2$, the difference co-array is composed of the set of self-differences $\hat{\mathbb{D}}_S = \hat{\mathbb{D}}_{11} \cup \hat{\mathbb{D}}_{22}$ and the set of cross-differences $\hat{\mathbb{D}}_C = \hat{\mathbb{D}}_{12} \cup \hat{\mathbb{D}}_{21}$, where $\hat{\mathbb{D}}_{ij}$ denotes the difference between $\hat{\mathbb{V}}_i$ and $\hat{\mathbb{V}}_j$. The difference co-array of $\hat{\mathbb{V}}_1 \cup \hat{\mathbb{V}}_2$ is expressed as

$$\begin{aligned} \hat{\mathbb{D}} &= \hat{\mathbb{D}}_S \cup \hat{\mathbb{D}}_C \\ &= \hat{\mathbb{D}}_{11} \cup \hat{\mathbb{D}}_{22} \cup \hat{\mathbb{D}}_{12} \cup \hat{\mathbb{D}}_{21} \\ &= (\hat{\mathbb{V}}_1 - \hat{\mathbb{V}}_1) \cup (\hat{\mathbb{V}}_2 - \hat{\mathbb{V}}_2) \cup (\hat{\mathbb{V}}_1 - \hat{\mathbb{V}}_2) \cup (\hat{\mathbb{V}}_2 - \hat{\mathbb{V}}_1), \end{aligned} \quad (12)$$

where $\hat{\mathbb{D}}_{11} = \{m'N \mid m' \in [0, M]\}$, $\hat{\mathbb{D}}_{22} = \{n'M \mid n' \in [0, N]\}$, $\hat{\mathbb{D}}_{12} = \{(m'N - n'M) \mid m' \in [0, M], n' \in [0, N]\}$, and $\hat{\mathbb{D}}_{21} = \{(n'M - m'N) \mid n' \in [0, N], m' \in [0, M]\}$.

Similarly, the difference co-array of $\mathbb{V}_1 \cup \mathbb{V}_2$ is

$$\begin{aligned} \mathbb{D} &= \mathbb{D}_S \cup \mathbb{D}_C \\ &= \mathbb{D}_{11} \cup \mathbb{D}_{22} \cup \mathbb{D}_{12} \cup \mathbb{D}_{21} \\ &= (\mathbb{V}_1 - \mathbb{V}_1) \cup (\mathbb{V}_2 - \mathbb{V}_2) \cup (\mathbb{V}_1 - \mathbb{V}_2) \cup (\mathbb{V}_2 - \mathbb{V}_1), \end{aligned} \quad (13)$$

where $\mathbb{D}_{11} = \{m'N \mid m' \in [0, M]\}$, $\mathbb{D}_{22} = \{n''M \mid n'' \in [0, N] \setminus \{N-1\}\}$, $\mathbb{D}_{12} = \{(m'N - n''M) \mid m' \in [0, M], n'' \in [0, N] \setminus \{N-1\}\}$, and $\mathbb{D}_{21} = \{(n''M - m'N) \mid n'' \in [0, N] \setminus \{N-1\}, m' \in [0, M]\}$.

$[0, N] \setminus \{N-1\}$, and $\mathbb{D}_{21} = \{(n''M - m'N) \mid n'' \in [0, N] \setminus \{N-1\}, m' \in [0, M]\}$.

It is straightforward to observe that $\hat{\mathbb{D}}_{11} = \mathbb{D}_{11}$. Owing to the symmetry of the difference co-array, only the positive part is analyzed. The self difference co-array between the removed sensor at $MN - M$ and the remaining sensors are given by

$$\begin{aligned} & \mathbb{D}_{MN-M|iM} \\ &= \{(MN - M) - iM \mid i \in [0, N-1] \setminus \{N-2\}\} \\ &= \{MN - jM \mid j \in [1, N] \setminus \{N-1\}\} \\ &= \mathbb{D}_{MN|jM} \in \mathbb{D}_{22}, \\ & \mathbb{D}_{MN-M|(N-2)M} \\ &= \{(MN - M) - (N-2)M\} \\ &= \{M - 0\} \in \mathbb{D}_{22}, \end{aligned} \quad (14)$$

This indicates that removing the sensor at $MN - M$ does not affect the self difference co-array, and hence $\hat{\mathbb{D}}_{22} = \mathbb{D}_{22}$.

The difference co-array between $MN - M$ and $\mathbb{V}_1$ is

$$\begin{aligned} & \mathbb{D}_{MN-M|iN} \\ &= \{(MN - M) - iN \mid i \in [0, M]\} \\ &= \{jN - M \mid j \in [0, M]\} \\ &= \mathbb{D}_{jN|M}, \end{aligned} \quad (15)$$

Therefore, removing the sensor at $MN - M$ does not affect the cross difference co-array, and hence $\hat{\mathbb{D}}_{12} \cup \hat{\mathbb{D}}_{21} = \mathbb{D}_{12} \cup \mathbb{D}_{21}$, leading to $\hat{\mathbb{D}} = \mathbb{D}$. Consequently, $\hat{\mathbb{V}}_1 \cup \hat{\mathbb{V}}_2$ and $\mathbb{V}_1 \cup \mathbb{V}_2$ share the same difference co-array.

Secondly, the holes in the difference co-array between $MN - M$ and subarray $\mathbb{P}_3$ will be filled by those generated between $MN + N$ and $\mathbb{P}_3$. The difference co-array between $MN - M$ and $\mathbb{P}_3$ is

$$\begin{aligned} & \mathbb{D}_{MN-M|\mathbb{P}_3} \\ &= \{(MN - M) - [-q(M + N)] \mid q \in [1, Q]\} \\ &= \{(MN + N) - [-q'(M + N)] \mid q' \in [0, Q-1]\} \\ &= \mathbb{D}_{MN+N|(\mathbb{P}_3 \cup 0)}, \end{aligned} \quad (16)$$

Based on these two-step analysis, the consecutive segments in the difference co-arrays of SACA and ACA are the same, and thus the same number of uDOFs can be provided.

B. Weight Function Analysis

For the weight functions of SACA, we need to analyze the impact of the shifted sensor from $MN - M$ to $MN + N$.

Proposition 2: For an SACA denoted by $\mathbb{P}_{\text{SACA}} = \mathbb{P}_1 \cup \mathbb{P}_2 \cup \mathbb{P}_3 \cup \mathbb{P}_4$, the weight functions with three nearest distance are given as follows.

For $M > 3$, we have

$$\begin{cases} w(1) = \begin{cases} 1, & N - M = 1, \\ 2, & \text{otherwise}, \end{cases} \\ w(2) = \begin{cases} 1, & N - M = 2, \\ 2, & \text{otherwise}, \end{cases} \\ w(3) = \begin{cases} 1, & N - M = 3, \\ 2, & \text{otherwise}, \end{cases} \end{cases} \quad (17)$$

For $M = 3$,

$$\begin{cases} w(1) = \begin{cases} 1, & N - M = 1, \\ 2, & \text{otherwise}, \end{cases} \\ w(2) = \begin{cases} 1, & N - M = 2, \\ 2, & \text{otherwise}, \end{cases} \\ w(3) = N - 2, \end{cases} \quad (18)$$

For $M = 2$,

$$\begin{cases} w(1) = \begin{cases} 1, & N - M = 1, \\ 2, & \text{otherwise}, \end{cases} \\ w(2) = N - 2, \\ w(3) = \begin{cases} 1, & N - M = 3, \\ 2, & N - M > 3, \\ 3, & N - M = 1. \end{cases} \end{cases} \quad (19)$$

Proof: We first present the proof for the case $M > 3$. For $M > 3$, the first three critical weight functions of ACA are $w(1) = 2$, $w(2) = 2$, and $w(3) = 2$ [24].

Firstly, we consider the impact of the newly added sensor at position $MN + N$. For $M > 3$ and given that $N > M > 3$, the distance between the added sensor at $MN + N$ and the sensor at MN is always larger than 3. Consequently, add a sensor at $MN + N$ has no impact on these critical weight functions.

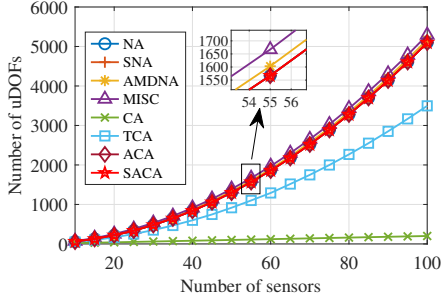
Secondly, the impact of removed sensor on the weight functions is examined. The adjacent sensors near $MN - M$ in $\mathbb{P}_1$ and $\mathbb{P}_2$ are $MN - 2N$, $MN - N$, $MN - M$, and MN . Denote D_1 as the distance between $MN - M$ and MN , D_2 represents the distance between $MN - M$ and $MN - N$, $D_3 = |(MN - M) - (MN - 2M)|$, while $D_4 = |(MN - M) - (MN - 2N)|$.

When $N - M = 1$, $D_1 = M > 3$, $D_2 = N - M = 1$, $D_3 = 2M - M = M > 3$. Since $N > M$, it follows that $D_4 = 2N - M > 2M - M = M > 3$. At this stage, removing the sensor at $MN - M$ results in a reduction of one in $w(1)$ compared with ACA, leading to $w(1) = 1$, $w(2) = 2$, and $w(3) = 2$ in the proposed SACA.

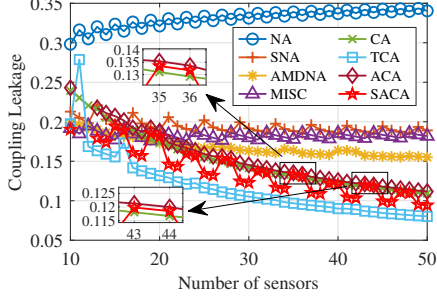
When $N - M = 2$, we have $D_1 = M > 3$, $D_2 = N - M = 2$, $D_3 = 2M - M = M > 3$, and $D_4 = 2N - M > 2M - M = M > 3$. Since $D_2 = 2$, removing the sensor at $MN - M$ reduces $w(2)$ compared to ACA. Consequently, in the SACA, $w(1) = 2$, $w(2) = 1$, and $w(3) = 2$.

When $N - M = 3$, $D_1 = M > 3$, $D_2 = N - M = 3$, $D_3 = 2M - M = M > 3$, and $D_4 = 2N - M > 2M - M = M > 3$. The distance between sensors $MN - M$ and $MN - N$ is 3. Therefore, we have $w(1) = 2$, $w(2) = 2$, and $w(3) = 1$ in SACA.

Finally, when $N - M > 3$, $D_1 = M > 3$, $D_2 = N - M > 3$, $D_3 = 2M - M = M > 3$, and $D_4 = 2N - M > 2M - M = M > 3$. Removing the array sensor at $MN - M$ does not affect the first three critical weight functions compared to ACA, and we can obtain that $w(1) = 2$, $w(2) = 2$, and $w(3) = 2$ in SACA.



(a) Number of uDOFs versus sensor number.



(b) Coupling leakage versus sensor number.

Fig. 2: Comparisons of uDOFs and coupling leakage.

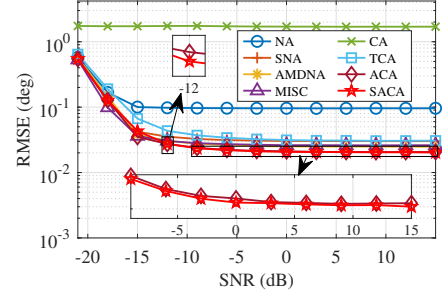
The above analysis completes the proof of the weight functions for the case $M > 3$. For cases with $M = 3$ and $M = 2$, the analysis procedure is similar and is omitted here. ■

IV. SIMULATION RESULTS

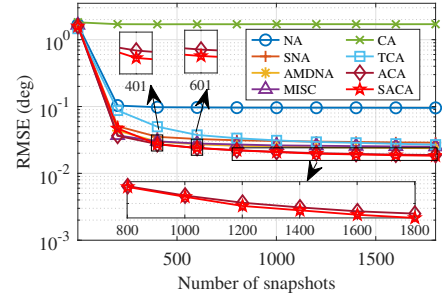
In this section, we compare the DOA estimation performance of the proposed SACA, NA [14], CA [15], SNA [21], AMDNA [22], TCA [23], ACA [24], and MISC [25]. We set $B = 100$ with coupling coefficients $c_1 = 0.3e^{j\pi/3}$, $c_l = c_1 e^{-j(l-1)\pi/8}/l$ for $l \in [2, B]$.

For the first set of simulations, the number of uDOFs and coupling leakage versus sensor number are shown in Fig. 2. It is noted that MISC provides the highest number of uDOFs, followed by AMDNA. Under the same sensor number, SACA offers the same number of uDOFs as that of ACA, NA, and SNA, while the number of uDOFs of CA is the smallest. For the mutual coupling, the coupling leakage of NA is the highest of all due to the largest $w(1)$ caused by the dense subarray configuration. TCA holds the smallest coupling leakage; however, its number of uDOFs is limited.

Denote T as the sensor number, and consider the values of M and N that achieve the maximum uDOFs. If T is even, i.e., $T = 2k$, $M + N = T/2 = k$, if T is odd, i.e., $T = 2k - 1$, $M + N = (T + 1)/2 = k$. There exists a combination of M and N satisfying both $M + N = k$ and $N - M = 1$ for odd k , leading to a smaller $w(1)$ which is translated to a significantly reduced coupling leakage. Therefore, the coupling leakage of SACA fluctuated with k and can be extremely better compared with ACA if an appropriate k is chosen.



(a) RMSE versus SNR.



(b) RMSE versus the number of snapshots.

Fig. 3: RMSE performance comparisons of DOA estimation for distinct sparse array geometries.

For the second set of simulations, the root mean square error (RMSE) is adopted as a metric for performance comparison, and SS-MUSIC [27] is employed for DOA estimation. The number of sensors is set to $T = 25$, with $K = 23$ sources uniformly distributed within $[-60^\circ, 60^\circ]$. The number of Monte Carlo trials is 1000. Fig. 3(a) depicts the relationship between RMSE and SNR with the number of snapshots fixed at $L = 1000$, and it can be concluded that the proposed SACA outperforms all other array configurations when SNR is greater than or equal to -12 dB. Fig. 3(b) shows the relationship between RMSE and the number of snapshots with SNR fixed at 0 dB, and the proposed SACA outperforms all other array configurations when the number of snapshots is greater than or equal to 401.

V. CONCLUSION

In this paper, a sparse array design, referred to as the shift-based augmented coprime array (SACA), was proposed. Compared to ACA, SACA achieves the same number of uDOFs. Furthermore, depending on the values of M and N , the first three critical weight functions in SACA can be reduced, thereby improving robustness against MC effect. It has been shown by simulations that the proposed structure achieves a more favorable balance between uDOFs and low MC effect, and better estimation performance is achieved with a fixed number of physical sensors.

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